Trabajo Práctico - Instalación de Docker y Despliegue de una Aplicación "Web Hello World"

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Parte 1 - Instalación de docker

1. Como docker no se encuentra en los repositorios de fedora, voy a instalar las herramientas necesarias para gestionar repositorios externos:

```
dani@fedora:~$ sudo dnf install dnf-plugins-core
Actualizando y cargando repositorios:
 Fedora 43 - x86_64 - Updates
Repositorios cargados.
Paquete                                                  Arq.          Versión
Instalando:
 dnf-plugins-core                                        noarch        4.10.1-6.fc43
Instalando dependencias:
 python3-dnf-plugins-core                                noarch        4.10.1-6.fc43

Resumen de la transacción:
 Instalando:          2 paquetes

El tamaño total de paquetes entrantes es 352 KiB. Se necesita descargar 352 KiB.
Después de esta operación, 1 MiB extra serán utilizados (instalar 1 MiB, eliminar 0 B).
Is this ok [y/N]: y
[1/2] dnf-plugins-core-0:4.10.1-6.fc43.noarch
[2/2] python3-dnf-plugins-core-0:4.10.1-6.fc43.noarch
--------------------------------------------------------------------------------
[2/2] Total
Ejecutando transacción
[1/4] Verificar archivos de paquete
[2/4] Preparar transacción
[3/4] Instalando python3-dnf-plugins-core-0:4.10.1-6.fc43.noarch
[4/4] Instalando dnf-plugins-core-0:4.10.1-6.fc43.noarch
¡Completado!
```

2. Descargando el repositorio oficial:

```
dani@fedora:~$ sudo dnf config-manager addrepo --from-repofile=https://download.docker.com/linux/fedora/docker-ce.repo
 https://download.docker.com/linux/fedora/docker-ce.repo
dani@fedora:~$
```

```
dani@fedora:~$ sudo dnf config-manager addrepo --from-repofile=
[https://download.docker.com/linux/fedora/docker-ce.repo]
(https://download.docker.com/linux/fedora/docker-ce.repo)

https://download.docker.com/linux/fedora/docker-ce.repo          100% |   4.4 KiB/s | 811.0   B |
00m00s

dani@fedora:~$
```

3. instalo docker desde el repositorio oficial:

```
dani@fedora:~$ sudo dnf install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Actualizando y cargando repositorios:
 Docker CE Stable - x86_64
Repositorios cargados.
Paquete                                                  Arq.          Versión
Instalando:
 containerd.io                                           x86_64        2.2.1-1.fc43
   reemplazando runc                                      x86_64        2:1.4.0-1.fc43
 docker-buildx-plugin                                     x86_64        0.30.1-1.fc43
 docker-ce                                                x86_64        3:29.1.5-1.fc43
 docker-ce-cli                                            x86_64        1:29.1.5-1.fc43
 docker-compose-plugin                                    x86_64        5.0.1-1.fc43
Instalando dependencias:
 libcgroup                                                x86_64        3.0-9.fc43
Instalando dependencias débiles:
 docker-ce-rootless-extras                                x86_64        29.1.5-1.fc43

Resumen de la transacción:
 Instalando:          7 paquetes
 Reemplazando:        1 paquete
```

```
El tamaño total de paquetes entrantes es 94 MiB. Se necesita descargar 94 MiB.
Después de esta operación, 354 MiB extra serán utilizados (instalar 367 MiB, eliminar 13 MiB).
Is this ok [y/N]: y
[1/7] docker-ce-cli-1:29.1.5-1.fc43.x86_64
[2/7] containerd.io-0:2.2.1-1.fc43.x86_64
[3/7] docker-ce-3:29.1.5-1.fc43.x86_64
[4/7] libcgroup-0:3.0-9.fc43.x86_64
[5/7] docker-compose-plugin-0:5.0.1-1.fc43.x86_64
[6/7] docker-buildx-plugin-0:0.30.1-1.fc43.x86_64
[7/7] docker-ce-rootless-extras-0:29.1.5-1.fc43.x86_64
--------------------------------------------------------------------------------
[7/7] Total
[1/8] https://download.docker.com/linux/fedora/gpg
[1/8] https://download.docker.com/linux/fedora/gpg
--------------------------------------------------------------------------------
[8/8] Total
Importando llave OpenPGP 0x621E9F35:
 IDUsuario     : "Docker Release (CE rpm) <docker@docker.com>"
 Huella dactilar: 060A61C51B558A7F742B77AAC52FEB6B621E9F35
 Desde                    : https://download.docker.com/linux/fedora/gpg
Is this ok [y/N]: y
La llave fue importada exitosamente.
[ 1/10] Verificar archivos de paquete
[ 2/10] Preparar transacción
[ 3/10] Instalando libcgroup-0:3.0-9.fc43.x86_64
[ 4/10] Instalando containerd.io-0:2.2.1-1.fc43.x86_64
[ 5/10] Instalando docker-ce-cli-1:29.1.5-1.fc43.x86_64
[ 6/10] Instalando docker-ce-3:29.1.5-1.fc43.x86_64
[ 7/10] Instalando docker-ce-rootless-extras-0:29.1.5-1.fc43.x86_64
[ 8/10] Instalando docker-compose-plugin-0:5.0.1-1.fc43.x86_64
[ 9/10] Instalando docker-buildx-plugin-0:0.30.1-1.fc43.x86_64
[10/10] Eliminando runc-2:1.4.0-1.fc43.x86_64
¡Completado!
dani@fedora:~$
```

```
dani@fedora:~$ sudo dnf install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

4. Levanto el servicio de docker:

```
dani@fedora:~$ sudo systemctl enable --now docker
[sudo] contraseña para dani:
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' → '/usr/lib/systemd/system/docker.service'.
dani@fedora:~$
```

```
dani@fedora:~$ sudo systemctl enable --now docker
[sudo] contraseña para dani:
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' →
'/usr/lib/systemd/system/docker.service'.

dani@fedora:~$
```

5. Pruebo el funcionamiento de docker:

```
dani@fedora:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:05813aedc15fb7b4d732e1be879d3252c1c9c25d885824f6295cab4538cb85cd
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

dani@fedora:~$
```

6. Me añado al grupo de docker para evitar usar sudo.

```
dani@fedora:~$ sudo usermod -aG docker $USER
dani@fedora:~$
```

7. Reinicio el servidor:

```
dani@fedora:~$ sudo shutdown -r now

Broadcast message from root@fedora on pts/2 (Mon 2026-01-19 08:25:59 -03):

The system will reboot now!

dani@fedora:~$ Read from remote host 192.168.1.6: Connection reset by peer
Connection to 192.168.1.6 closed.
client_loop: send disconnect: Broken pipe
dani@fedora:~$
```

Parte 2 - Comandos básicos de docker

Aquí voy a probar los comandos básicos de docker.

- docker version:

```
dani@fedora:~$ docker version
Client: Docker Engine - Community
 Version:           29.1.5
 API version:       1.52
 Go version:        go1.25.6
 Git commit:        0e6fee6
 Built:             Fri Jan 16 12:51:41 2026
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Engine - Community
 Engine:
  Version:          29.1.5
  API version:      1.52 (minimum version 1.44)
  Go version:       go1.25.6
  Git commit:       3b01d64
  Built:            Fri Jan 16 12:48:28 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
dani@fedora:~$
```

- docker info:

```
dani@fedora:~$ docker info
Client: Docker Engine - Community
 Version:    29.1.5
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.30.1
    Path:     /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.0.1
    Path:     /usr/libexec/docker/cli-plugins/docker-compose

Server:
 Containers: 2
  Running: 0
  Paused: 0
  Stopped: 2
 Images: 1
 Server Version: 29.1.5
 Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
 Logging Driver: json-file
 Cgroup Driver: systemd
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 runc
```

```
Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version: dea7da592f5d1d2b7755e3a161be07f43fad8f75
runc version: v1.3.4-0-gd6d73eb8
init version: de40ad0
Security Options:
 seccomp
  Profile: builtin
 cgroupns
Kernel Version: 6.18.5-200.fc43.x86_64
Operating System: Fedora Linux 43 (Server Edition)
OSType: linux
Architecture: x86_64
CPUs: 2
Total Memory: 3.031GiB
Name: fedora
ID: fb36cfa1-6537-4e96-9530-6b0da4238969
Docker Root Dir: /var/lib/docker
Debug Mode: false
Experimental: false
Insecure Registries:
  ::1/128
  127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables+firewalld

dani@fedora:~$
```

- docker pull hello-world

```
dani@fedora:~$ docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
dani@fedora:~$
```

- docker images

```
dani@fedora:~$ docker images
                                                    i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   ef54e839ef54       21.8kB         9.52kB    U
dani@fedora:~$
```

- docker run hello-world

```
dani@fedora:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

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 $ docker run -it ubuntu bash

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 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

dani@fedora:~$
```

- docker ps -a

```
dani@fedora:~$ docker ps -a
CONTAINER ID   IMAGE          COMMAND    CREATED          STATUS                      PORTS     NAMES
98232771fe3d   hello-world    "/hello"   12 minutes ago   Exited (0) 12 minutes ago             naughty_greider
39c653301fad   05813aedc15f   "/hello"   3 weeks ago      Exited (0) 3 weeks ago                quirky_lumiere
7fe2164b9461   05813aedc15f   "/hello"   3 weeks ago      Exited (0) 3 weeks ago                tender_herschel
dani@fedora:~$
```

Parte 3 - Crear una Aplicación Web Simple

1. Creo el directorio del proyecto e ingreso para trabajar en el:

```
dani@fedora:~$ mkdir docker-webapp && cd docker-webapp
dani@fedora:~/docker-webapp$
```

2. Creo el index.html de la aplicación web:

```
dani@fedora:~/docker-webapp$ echo > index.html
```

3. Edito el index.html:

```
dani@fedora:~/docker-webapp$ nano index.html
```

```
  GNU nano 8.5                                                    index.html
<!DOCTYPE html>
<html>
<head>
        <title>¡Hola ISO!</title>
</head>
<body>
        <h1>Hola mundo desde Docker</h1>
        <p>Esta es mi primera aplicación web en un contenedor Docker.</p>
</body>
</html>

                                                   [ 10 líneas escritas ]
^G Ayuda      ^O Guardar     ^F Buscar      ^K Cortar      ^T Ejecutar    ^C Ubicación
^X Salir      ^R Leer fich.  ^\ Reemplazar  ^U Pegar       ^J Justificar  ^/ Ir a línea
```

4. Creo el Dockerfile:

```
dani@fedora:~/docker-webapp$ echo > dockerfile
dani@fedora:~/docker-webapp$ nano dockerfile
```

```
  GNU nano 8.5                                                    dockerfile
# Usar una imagen base de Nginx
FROM nginx:alpine

# Copiar nuestro archivo HTML al directorio de Nginx
COPY index.html /usr/share/nginx/html/index.html

# Exponer el puerto 80
EXPOSE 80

# Comando para iniciar Nginx
CMD ["nginx", "-g", "daemon off;"]

^G Ayuda      ^O Guardar     ^F Buscar      ^K Cortar      ^T Ejecutar    ^C Ubicación
^X Salir      ^R Leer fich.  ^\ Reemplazar  ^U Pegar       ^J Justificar  ^/ Ir a línea
```

Parte 4 - Construir y Ejecutar el Contenedor

1. Construyo la imagen:

```
dani@fedora:~/docker-webapp$ docker build -t my-webapp-iso .
[+] Building 17.4s (7/7) FINISHED
 => [internal] load build definition from dockerfile
 => => transferring dockerfile: 350B
 => [internal] load metadata for docker.io/library/nginx:alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 289B
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
 => => resolve docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
 => => sha256:5e7756927bef33a266e1221356d5da8553139cb80bc5b1b3827010811d9ea268 20.24MB / 20.24MB
 => => sha256:955a8478f9aceb66cbf2f579fa3c24e1af278d1fa3ffd3043d6260e21d2f7416 1.21kB / 1.21kB
 => => sha256:6d397a54a185dd0b6638d1a3934b81daef7a140741e12697377d6279066f7ca7 1.40kB / 1.40kB
 => => sha256:399d0898a94e0084f81499a3e3c29824357118c7ce551648ea1dbab813884661 404B / 404B
 => => sha256:6b7b6c7061b76cdb8601e18722d12ae3232f0ddcfa1d2983754abcc6ce0a8a83 954B / 954B
 => => sha256:bca5d04786e112d958f100a66f8257b2aeefc14b64d81e405c3c44acff2fb000 1.86MB / 1.86MB
 => => sha256:3e2c181db1b0985ce357c7aaf48ac615f30f392cd15d5c5ba34c4faa1f4f39a2 626B / 626B
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153
 => => extracting sha256:bca5d04786e112d958f100a66f8257b2aeefc14b64d81e405c3c44acff2fb000
 => => extracting sha256:3e2c181db1b0985ce357c7aaf48ac615f30f392cd15d5c5ba34c4faa1f4f39a2
 => => extracting sha256:6b7b6c7061b76cdb8601e18722d12ae3232f0ddcfa1d2983754abcc6ce0a8a83
 => => extracting sha256:399d0898a94e0084f81499a3e3c29824357118c7ce551648ea1dbab813884661
 => => extracting sha256:955a8478f9aceb66cbf2f579fa3c24e1af278d1fa3ffd3043d6260e21d2f7416
 => => extracting sha256:6d397a54a185dd0b6638d1a3934b81daef7a140741e12697377d6279066f7ca7
 => => extracting sha256:5e7756927bef33a266e1221356d5da8553139cb80bc5b1b3827010811d9ea268
 => [2/2] COPY index.html /usr/share/nginx/html/index.html
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:803e87fa709ce060dbc94ed4e39977243e23ab332f7da8137d6bc235b7f4bb1c
 => => exporting config sha256:449033bc57d0b6c53951f679291fd8029fcaac4e1ab12688de1554b02d11a85e
 => => exporting attestation manifest sha256:89410c37c6a123a1cff5a7b89dbf901cb8634c4b6467eb56a3de088a9a9e4d9e
 => => exporting manifest list sha256:94d82d451a615942090af1f79d0d37fee8eb84e00df0c4c8b488e273c87a8ad3
 => => naming to docker.io/library/my-webapp-iso:latest
 => => unpacking to docker.io/library/my-webapp-iso:latest
dani@fedora:~/docker-webapp$
```

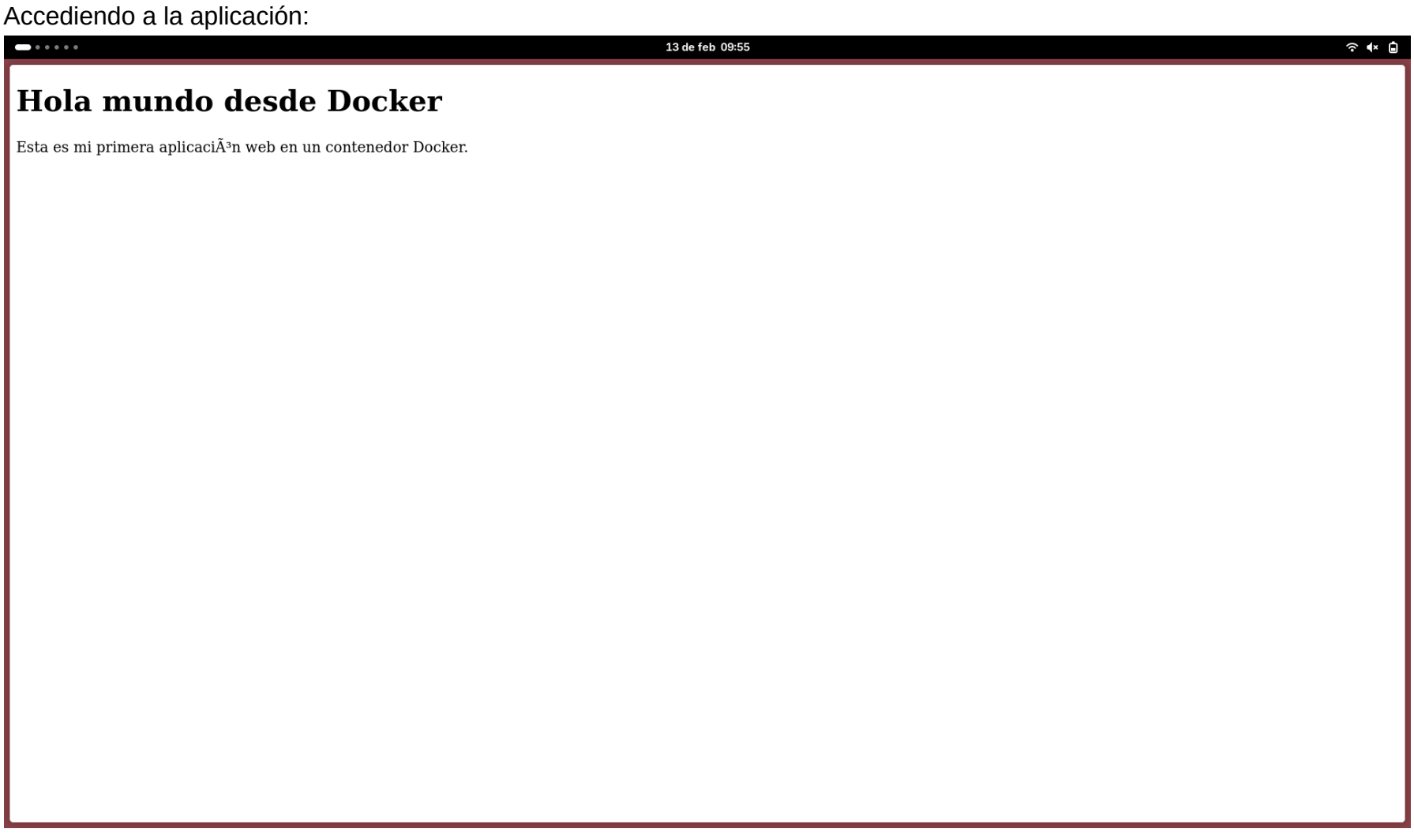
2. Ejecuto el contenedor:

```
dani@fedora:~/docker-webapp$ docker run -d -p 8080:80 --name webapp-iso my-webapp-iso
b054434fde0ea073e45ed60ae536d945a493786a851c39df64d54f171b4c3847
dani@fedora:~/docker-webapp$
```

3. Verifico si se levanto el contenedor:

```
dani@fedora:~/docker-webapp$ docker ps
CONTAINER ID   IMAGE           COMMAND                  CREATED         STATUS         PORTS                                       NAMES
b054434fde0e   my-webapp-iso   "/docker-entrypoint.…"   2 minutes ago   Up 2 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   webapp-iso
dani@fedora:~/docker-webapp$
```

4. Accediendo a la aplicación:



Parte 5 - Tareas Adicionales (Opcionales)

Modifico el contenido del index.html

1. Paro el contenedor:

```
dani@fedora:~/docker-webapp$ docker stop webapp-iso
webapp-iso
dani@fedora:~/docker-webapp$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
dani@fedora:~/docker-webapp$
```

2. Modifico el index.html:

```
dani@fedora:~/docker-webapp$ nano index.html
  GNU nano 8.5                                                    index.html
<!DOCTYPE html>
<html>
<head>
        <title>¡Hola ISO!</title>
</head>
<body>
        <h1>Hola mundo desde Docker</h1>
        <p>Esta es mi primera aplicación web en un contenedor Docker.</p>
        <p>Y esta es la modificación<p>
</body>
</html>
```

3. Reconstruyo la imagen:

```
dani@fedora:~/docker-webapp$ docker build -t my-webapp-iso .
[+] Building 5.9s (7/7) FINISHED
 => [internal] load build definition from dockerfile
 => => transferring dockerfile: 350B
 => [internal] load metadata for docker.io/library/nginx:alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 323B
 => CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
 => => resolve docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
 => [2/2] COPY index.html /usr/share/nginx/html/index.html
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:d3673037cfa108b26d6e1874f9173211d0d7dcc2f2dd6ec1d7c0aa21b1dc24c1
 => => exporting config sha256:9649618a2db7d99cc3fdb1e6b8e170f335536c1a086e91d1e227d4733d446331
 => => exporting attestation manifest sha256:1a8208d7c41040c60ca21b6f2b92806f833f22688df75954bbc488056bb9af62
 => => exporting manifest list sha256:a79cde572240bcd8b414a0f55e321ab15ba641154ce081c9d64468799c3fe139
 => => naming to docker.io/library/my-webapp-iso:latest
 => => unpacking to docker.io/library/my-webapp-iso:latest
dani@fedora:~/docker-webapp$
```

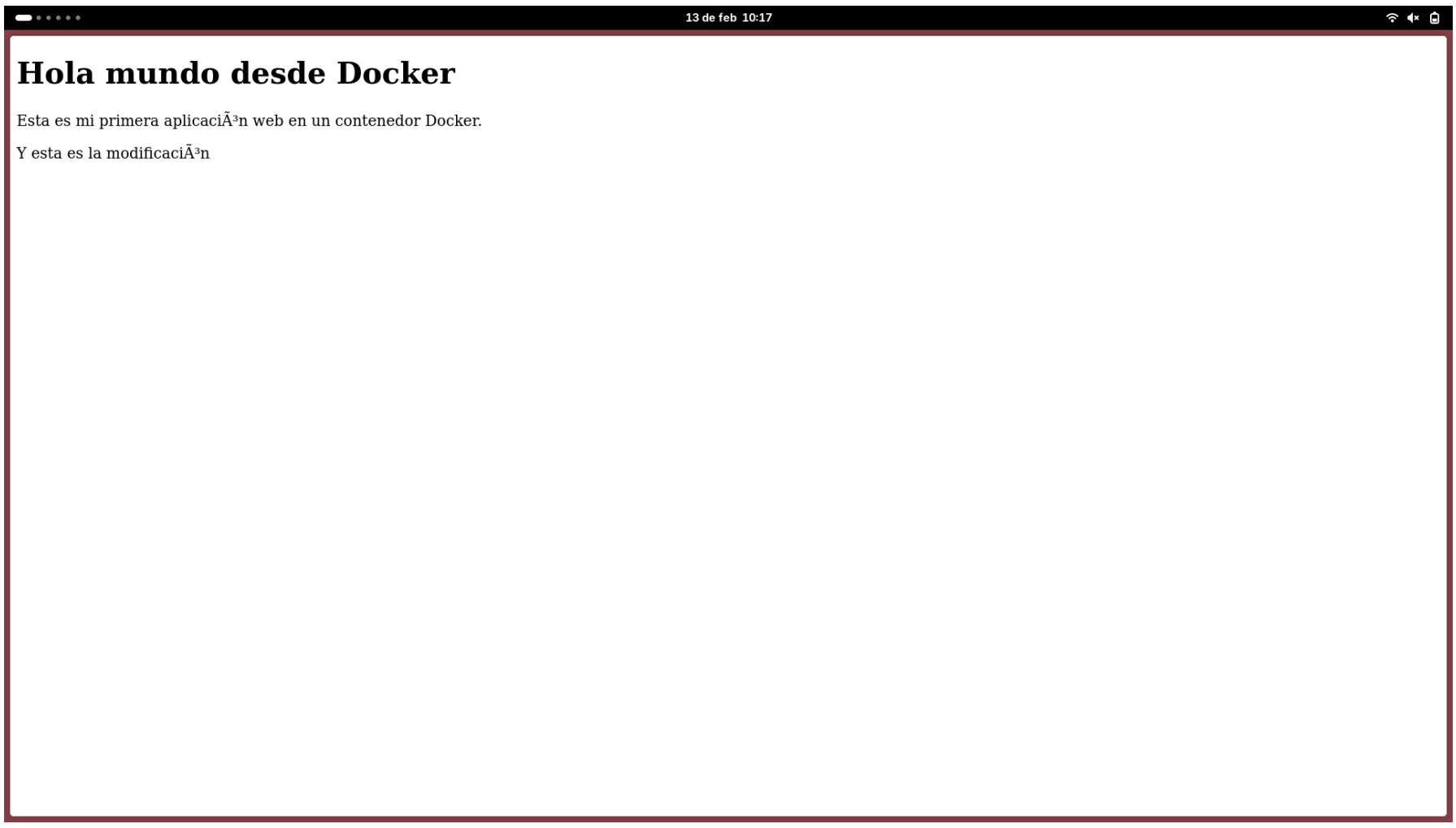
4. Borro el contenedor:

```
dani@fedora:~/docker-webapp$ docker rm webapp-iso
webapp-iso
dani@fedora:~/docker-webapp$
```

5. Levanto el contenedor con la modificación:

```
dani@fedora:~/docker-webapp$ docker run -d -p 8080:80 --name webapp-iso my-webapp-iso
8507bd555c8469f5d56a6ca7ab55b7306c56e527fa32a164986edb841126a017
dani@fedora:~/docker-webapp$
```

6. Verifico si se realizo la modificación:



Pruebas de docker stop y docker start:

```
dani@fedora:~/docker-webapp$ docker stop webapp-iso
webapp-iso
dani@fedora:~/docker-webapp$ docker start webapp-iso
webapp-iso
dani@fedora:~/docker-webapp$
```

Pruebas de borrado:

1. Elimino el contenedor:

```
dani@fedora:~/docker-webapp$ docker ps -a
CONTAINER ID   IMAGE           COMMAND                  CREATED         STATUS                     PORTS     NAMES
8507bd555c84   my-webapp-iso   "/docker-entrypoint.…"   8 minutes ago   Exited (0) 10 seconds ago            webapp-iso
dani@fedora:~/docker-webapp$  docker rm webapp-iso
webapp-iso
dani@fedora:~/docker-webapp$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
dani@fedora:~/docker-webapp$
```

2. Elimino la imagen:

```
dani@fedora:~/docker-webapp$ docker images

IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     ef54e839ef54       21.8kB         9.52kB
my-webapp-iso:latest   a79cde572240         91MB           26MB
nginx:latest           341bf0f3ce6c        237MB         65.8MB
dani@fedora:~/docker-webapp$ docker rmi my-webapp-iso:latest
Untagged: my-webapp-iso:latest
Deleted: sha256:a79cde572240bcd8b414a0f55e321ab15ba641154ce081c9d64468799c3fe139
dani@fedora:~/docker-webapp$ docker images

IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     ef54e839ef54       21.8kB         9.52kB
nginx:latest           341bf0f3ce6c        237MB         65.8MB
dani@fedora:~/docker-webapp$
```

Parte 6 - preguntas

1. Docker ofrece:
    1. Eficiencia en el uso de recursos: al usar el kernel del host y el userspace del S.O, se vuelve mucho mas ligero y eficiente.
    2. Velocidad de arranque: Docker ofrese un arranque muy veloz debido a que al no virtualizar todo un computador, la cantidad de procesos a ejecutar es la justa y necesaria.
    3. Portabilidad: La portabilidad de docker es mucho mayor que en las MV debido a la ligereza de sus contenedores y la forma de crear imágenes.

2.
    1. Linea 1: Define cual es la imagen base que se utilizara (Nginx).
    2. Linea 2: Copia el archivo index.html y lo pega en la ruta `/usr/share/nginx/html/index.html` del sistema de archivos del contenedor.
    3. Linea 3: Expone el puerto 80
    4. Linea 4: Ejecuta el comando `["nginx", "-g", "daemon off;"]` dentro del contenedor
3. Se crea un **túnel de red** entre tu máquina física (el host) y el contenedor, mapeando el puerto 8080 del contenedor al puerto 9090 del host.
4. Algunos de los usos que se le puede dar a docker es:
    1. Manejo múltiples aplicaciones en servidores.
    2. Despliegue y manejo de microservicios.
    3. Transporte de aplicaciones para diversos fines.
    4. Manejo simple y optimo del CD/CI.